



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - Executive brief

Drafted 08/10/2026

INVESTIGATION WORKFLOW SUITE

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What happened

In the early hours of 12 August 2026, a 220 t class rigid haul truck (HT-407) was working night shift in the Stage 4 open pit at Bindarra Gold Mine. The pit had returned from crib (the meal break) at 02:22. The truck was on its second cycle since then, coming back down the main ramp empty. At RL 312, a 240 m access road, or spur, branched off the ramp to a dewatering bore. The spur had been opened to light vehicles on 2 August 2026. On that day the construction sign at its mouth was removed and nothing was put in its place. On 4 August a dewatering technician reported a near miss with a descending haul truck at the same intersection. The light vehicle (LV-118) was driven by a grade control technician employed by a survey subcontractor. He was on his third night shift at the mine. At about 02:45 he finished his survey work at the bore and drove out along the spur, reaching its mouth at 02:47. Nobody else on the shift knew he was in the pit.

At 02:47:06 the truck was travelling at 37 km/h, 53 m above the spur mouth. The survey places the first sight of the light vehicle's flashing beacon over the crest at this point. Within 1.4 seconds the operator braked hard, applied the retarder (an additional braking system) at full and steered right. At 02:47:09 the truck's collision awareness system detected the light vehicle 21.4 m away. It gave no warning in the cab, because the spur mouth lay inside a mapped zone in which alerts were suppressed. At 02:47:12 the front left of the truck, slowed to 24 km/h, struck the right side of the light vehicle 6.2 m into the ramp. The light vehicle was pushed 18.4 m down the ramp and rolled onto its left side against the inner windrow (an earth bank along the road edge). The truck stopped 19 m below the intersection. Dispatch stopped all Stage 4 traffic at 02:48:01 and paged the emergency response team. The first response vehicle reached the scene at 02:54. The second responder brought the hydraulic rescue tool from the village 11 km away and arrived at 03:07. The driver was freed through the rear of the vehicle at 03:14. He was conscious throughout. He had a fractured left collarbone, three fractured left ribs and a small pneumothorax (air leaking around a lung), and he was flown to the metropolitan trauma centre at 07:40. The seatbelt carried loading marks and the roll-over protection structure was bent inward. The light vehicle was destroyed. The truck was damaged at the front left and returned to service on 14 August. The Stage 4 pit stood down for 9 hours 15 minutes, and the main ramp below RL 340 was closed for 22 hours.

Why it happened

The collision happened because neither driver knew the other was there until the truck came over the crest. It is extremely likely that this lack of awareness influenced the collision. Several conditions combined to produce it.

One was the layout of the intersection itself. At the spur mouth the sight line was 41 m, but a descending haul truck needed 96 m to stop. It is very likely that this shortfall contributed to neither driver seeing the other in time. Once the light vehicle came into view, the distance available was well short of what the truck needed to stop.

Two controls that could have warned the drivers despite the poor sight line were not in place. The first was a positive call [NOT ESTABLISHED: the site's definition of a positive call, for a plain-language explanation]. Neither driver made one before entering the intersection, and it is likely that this contributed. The reason lay in the site's rules. Positive calls were required only at intersections on a list, and new intersections were added to that list only at the annual review. It is very likely that this arrangement led to no call being made at the newly opened spur. Behind it sat a guideline that set no trigger for a review and no basis for deciding which intersections belonged on the list. It is likely that this gap in the guideline shaped the site's arrangement.

The second missing control was the collision awareness system. The spur mouth lay inside a geofence (a mapped zone) in which alerts were suppressed, and the operators knew nothing of it. It is likely that this contributed to neither driver being aware of the other.

At the same time, the light vehicle drove out of the spur onto the ramp without stopping. It is extremely likely that this contributed to the lack of awareness. Two conditions lay behind it. First, the intersection had no stop control or warning sign, even though it failed the sight distance rule. It is likely that this contributed. Second, the driver's access card had been issued with no check of the pit driving assessment, and it is likely that this also contributed.

The missing sign and the suppressed alerts shared a common origin. The spur was opened under a works order, so nothing about it was referred to anyone. It is very likely that this led to both the absence of stop control or a warning sign and the suppression of collision alerts at the spur.

The investigation identified six safety issues in this sequence:

- the guideline's lack of a review trigger and of a basis for the positive-call list
- the opening of the spur under a works order
- the issue of access cards without a check of the pit driving assessment
- positive calls being required only at listed intersections, with additions made only at the annual review
- the suppression of collision alerts by a geofence unknown to operators
- the absence of stop control or a warning sign at an intersection failing the sight distance rule

Two further safety issues were identified. These were the rostering of haul operators for fourteen consecutive 12.5 hour night shifts, and night shift extrication having to wait on a second responder called from the village. It is unlikely that either influenced this occurrence.

What we found

A finding stated without a likelihood term was established as fact.

Contributing safety factors

- The RL 312 spur was opened under a works order, so nothing was referred to anyone, and it is very likely that this contributed to the collision. (Safety issue)
- It is very likely that collision alerts were suppressed at the spur by a geofence (a mapped zone) operators knew nothing of, and it is likely that this contributed to the collision. (Safety issue)
- At the spur, a sight line of 41 m stood against a 96 m stopping distance, and this very likely contributed to the collision.
- The guideline (MSG-14) under which the site's traffic management plan was made sets no review trigger and no basis for the positive-call list, and it is likely that this contributed to the collision. (Safety issue)
- Positive calls were required only at listed intersections, which were added only at the annual review, and this arrangement very likely contributed to the collision. (Safety issue)
- Neither driver made a positive call before entering the intersection, which the plan did not require there, and it is likely that this contributed to the collision.
- Access cards were issued with no check of the pit driving assessment, and in all likelihood this contributed to the collision. (Safety issue)
- There was no stop control or warning sign at an intersection that failed the sight distance rule, and it is probable that this contributed to the collision. (Safety issue)
- The light vehicle drove out of the spur onto the ramp without stopping, an action that very likely contributed to the collision.
- It is very likely that neither driver was aware of the other until the truck came over the crest, and very likely that this contributed to the collision.
- The haul truck's front left struck the right side of the light vehicle at 24 km/h, and the evidence is conclusive that this impact contributed to the driver's injuries.
- Pushed 18 m down the ramp, the light vehicle rolled onto its left side, and this was conclusively established as contributing to the driver's injuries.

Other factors that increased risk

- Haul operators were rostered for fourteen consecutive 12.5 hour night shifts. It is likely that this increased risk across the operation, although it did not influence this collision. (Safety issue)
- On night shift, extrication waited on a second responder called from the village, and it is very likely that this increased the risk to any person trapped after an occurrence. (Safety issue)

Other findings

- The investigation examined whether the ramp surface was wet or slippery at and above the intersection, and found this very unlikely.
- The possibility that the service brakes delivered less than their tested deceleration during the stop was also examined and found very unlikely.

- The truck operator braked and steered right within 1.4 seconds of the beacon appearing. This very likely reduced risk by lowering the impact speed to 24 km/h from the 37 km/h it would otherwise have been.

What is being done

The investigation issued a preliminary update on 28 August 2026. The action recorded against each safety issue at the time of writing is set out below.

Positive calls were required only at listed intersections, and new intersections were added to the list only at the annual review. On 12 August 2026, the mine issued a site-wide direction requiring a positive call at every Stage 4 intersection, whether listed or not. Before this report was drafted, the investigation raised the annual-review listing rule with the Mining Manager. [NOT ESTABLISHED: any change made or proposed to the listing rule in the traffic management plan, and its status]

The guideline (MSG-14) sets no review trigger and gives no basis for the positive-call list. No action has been recorded against this issue.

Because the spur was opened under a works order, it was never referred to the traffic management plan's owner, the survey department or the technical services group. No action has been recorded against the works order exclusion in the mine's management of change procedure.

The RL 312 intersection failed the sight distance rule, yet it had no stop control or warning sign. No action has been recorded. [NOT ESTABLISHED: whether controls have since been installed at the intersection]

At the spur, collision alerts were suppressed by a geofence that operators did not know about. No action has been recorded against the geofence boundary, its review or operator training.

Access cards were issued without any check of the pit driving assessment. No action has been recorded against the gatehouse process.

Haul operators were rostered for fourteen consecutive night shifts of 12.5 hours each. No action has been recorded against this issue.

On night shift, extrication had to wait for a second responder called from the village. The site's own post-incident review measured that delay at approximately 9 minutes. The investigation has asked the emergency response coordinator how responders are rostered across the site's other three night shifts, and the issue remains open until that answer is received. No change to the arrangement has been recorded.

Distribution

Title	Name
General Manager, Bindarra Gold Mine	T. Brandner
Registered Manager (statutory)	S. Kirkhope
Mining Manager and owner of BIN-TMP-004	H. Quaile
Manager Health and Safety, Orrabinda Resources	L. Farquhar
Project Manager, Tarrown Mining Services	D. Mokoena
Superintendent Dewatering, Orrabinda Resources	P. Nyaguse